# 3D laser scanning, as-built models and the engineering that follows

We scan the hull or structure, build the as-built 3D model, and run the structural and stability work on that same model. All in one small firm in Newport, RI.

| UEI | CAGE | BUSINESS TYPE | NAICS |
|---|---|---|---|
| NWJRCE69AWH9 | Pending | Small business · Since 2018 | 541330 · 541370 · 541380 · 541690 · 541715 |

## What we do

- **3D laser scanning.** Hulls, decks, appendages and structure, scanned with our own FARO scanner.
- **As-built 3D models.** Surface models and drawings in Rhino and SolidWorks, ready for engineering, repair or new parts.
- **Structural analysis on the scan.** FEA on the as-built geometry in Femap/Nastran and SolidWorks Simulation.
- **Stability and USCG compliance.** Hydrostatics and stability in GHS and Orca3D, inclining experiments, and stability and rig-loads reports addressed to the USCG (46 CFR Subchapter T).
- **Deliverables.** Registered point cloud in E57 and native FARO format, a FARO SCENE registration report, the as-built model in Rhino or SolidWorks, and an engineering report when analysis is ordered.

## Why Wyspa

- Scan, model and analysis in one firm. No hand-off between a scan vendor and an engineering firm.
- Naval architects run the scan, so the model comes back ready for engineering.
- We come to the vessel. Recent scans were done on site in a working repair yard. We handled yard access and insurance paperwork.

**101 of 191**
projects where scanning fed the engineering

**85 ft**
aluminum patrol boat, scanned on site at a repair yard (2026)

## Track record (commercial)

- **85 ft aluminum patrol boat** (2026). Scanned on site in a working repair yard; as-built 3D model of the outer surfaces in progress.
- **USCG Subchapter T conversions.** A historic 12 Metre (USCG approved) and a 12-passenger Pearson 35 (structure approved, rig review pending). Scans on both; inclining test, stability and rig-loads reports on the Pearson.
- Other scans: a 114 ft motor yacht, an aircraft fuselage and wings, keels, rudders and rigs.

**Casey Brown**, Owner & Naval Architect
B.S. Naval Architecture & Marine Engineering, Webb Institute (2016)

casey@wyspa.tech · (401) 523-4920
79 Thames St, Newport, RI 02840 · gov.wyspa.tech